

ABDULRAHMAN SIRELKHATIM TAGALSIR ALI

Medical Graduate (MBBS) | Health Data Scientist | Organizational Leader
+249908115725 | abdurahmanalsir4@gmail.com | Omdurman, Sudan | [LinkedIn](#) | [Website](#)

PROFESSIONAL SUMMARY

Medical doctor and health data scientist combining 5+ years of clinical training with advanced data science skills (MIT Emerging Talent Program, IBM Data Science Professional Certificate, DeepLearning.AI Mathematics for Machine Learning). Delivered 28 complete data analysis projects for medical researchers across 10+ specialties. Built functional software including a mobile application and Windows desktop program from scratch. Led Sudan's largest medical students' association (800+ members) through war-time disruption. Seeking to apply clinical knowledge and analytical expertise to health data science research and practice.

EDUCATION

MBBS — Bachelor of Medicine and Surgery | Sudan University of Science and Technology

November 2018 – July 2026 | GPA 3.22/4.0

- Graduation Thesis: Knowledge, Attitudes, and Practices Study on Stroke Awareness Among Sudanese Population — Grade A. Cross-sectional survey with Python and SPSS analysis. Informs stroke prevention public health strategy in Sudan.
- Maintained progression through: 2019 Sudanese Revolution (10-month closure), COVID-19 (6-month closure), 2021 faculty strike, April 2023 armed conflict (ongoing displacement)
- Relevant coursework: Neuroscience (Central and Autonomic Nervous Systems), Neurology, Biostatistics, Epidemiology, Research Methods, Evidence-Based Medicine, Internal Medicine, Emergency Medicine

Certificate in Computer and Data Science | Massachusetts Institute of Technology — Emerging Talent Program

November 2024 – December 2025 | 5% acceptance rate

- Foundation Track: Python, Git and GitHub version control, discrete mathematics, computational thinking, debugging, software engineering practices
- Capstone: Humanitarian Crisis Intelligence and Communication System — geospatial platform fusing International Organization for Migration displacement tracking data with Humanitarian OpenStreetMap data to map internally displaced persons against health facility availability for Ministry of Health resource allocation. Built using Python, GeoPandas, Folium, and Streamlit. Repository on GitHub.

Pan African Youth Leadership Program | University of Texas at Austin — US Department of State

July 2017

- Competitive program for outstanding African youth leaders. Mini Diploma in Business Administration. Certificates from the University of Texas at Austin, US Department of State, and Meridian International Center.

PROFESSIONAL EXPERIENCE

Freelance Health Data Analyst | Self-Employed

October 2025 – Present | Remote

- 28 complete data analysis projects for MD, MBBS, Clinical Master, and Doctoral researchers across 10+ medical specialties including oncology, infectious disease, psychiatry, ophthalmology, maternal health, and public health
- Full-service support per project: data cleaning, descriptive and inferential statistics, multi-variable regression, survival analysis, ROC curves, publication-ready visualizations, results chapter writing, and thesis defense presentations. All 11 clients who presented successfully defended.
- Tools: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn, matplotlib, seaborn), SPSS, Excel. Full portfolio on GitHub.

Deputy Director, Resources and Excellence Management | Sudanese Consortium for Surgical Development

August 2024 – May 2026 | Remote

- Led 5+ recruitment cycles: reviewed 80+ applications, conducted 20-30 interviews, managed onboarding for 4 departments. Defined Key Performance Indicators for all positions from Chief Visionary Officer to entry-level.
- Developed annual monitoring and evaluation framework for all departments; built and maintained organizational website (HTML, CSS, PHP); drafted partnership guidelines and operational frameworks; covered Director duties for 5 weeks

LEADERSHIP EXPERIENCE

President | Sudan University Medical Students Association

March – December 2025 | 800+ students

- Rebuilt association from near-collapse: recruited executive committee members, filled 2 vacant positions, restored full operational capacity within 2 months
- Built organizational monitoring and evaluation mobile app (Flutter): term management, member database, 5-pillar weighted scoring, monthly and quarterly and annual evaluations, automated awards, progress charts
- Built Windows certificate generation program: HTML templates, Excel data import, QR code verification, unique certificate identifiers, bulk generation for entire organization
- MedXplore 2025: 7-day online orientation program for 465 students (56% of 833 total student body) across all 7 batches, aligned with CanMEDS Physician Competency Framework
- Sudan University Medical Students Association Professional Development Program: 16-session capacity building program attended by 216 members covering leadership, strategic planning, financial management, and organizational management
- Led constitutional reform through executive committee, council approval, and general assembly ratification (400+ voters). Headed Elections Committee: automated vote tabulation with Google Sheets script, results published within 24 hours.

Secretary General | *Sudan University Medical Students Association*

February – December 2023 | 7-member executive committee

- Directed all association operations as acting president during unprecedented armed conflict (April 2023): pivoted organization to academic stability support, maintained operations with zero unplanned downtime
- Established 2,800+ document digital archive from scratch: collected from 38+ email accounts, Telegram, Facebook, and cloud drives; classified into 13 categories; reduced retrieval from days to minutes
- Organized 36 meetings; submitted 8 monthly reports; issued 230+ certificates. Led fundraising committee: raised 12,000,000+ Sudanese Pounds (approximately \$5,000) supporting 98+ war-affected students.

External Affairs Secretary | *Sudan University Medical Students Association*

June 2022 – January 2023

- Drafted 371 letters to government entities, NGOs, private sector, and international organizations. Secured \$14,000+ in financial and in-kind support. Signed collaboration contracts with partner organizations.

Capacity Building Development Assistant — National Standing Committee on Professional Exchange | *Medical Students International Network – Sudan (MedSIN-Sudan)*

July – December 2021

- Founded Training New Exchange Facilitators workshop: increased certified facilitators from 7 to 20+, now the official national training program for the Standing Committee on Professional Exchange in Sudan
- Organized first national Standing Committee on Professional Exchange activity in 5 years (Global Health and Universal Health Coverage workshop). Drafted first-ever Capacity Building Guidelines. Named Assistant of the Term — 100% of objectives achieved.

Batch Coordinator | *SUST College of Medicine*

May 2022 – July 2026

- Served as primary liaison between 4-member coordination team and entire academic batch (130+ students)
- Facilitated communication between students, faculty, and administration
- Coordinated batch-wide academic support, activities, and addressing student concerns throughout final years of medical school

FIELD & CLINICAL EXPERIENCE

Group Head (Group B) | *SUMSA Medical Mission — Garri Area, Khartoum State*

April 2026 | Field Deployment

- Led Group B field deployment (32 students + 8 lab members + pharmacist + security) serving 600–650 patients across medical day operations in one of Khartoum's most war-affected areas
- Responsibilities: morning check-in, tag management, overseeing 6 offices (clinics, pharmacy, lab, research, health education, environmental health), anticipating and resolving real-time operational problems, daily team debrief, end-of-day logistics

Rural Health Trainee | *SUST Rural Health Training Programme — Um Katti Hospital*

April 2026 | Karari Locality, Khartoum, Sudan

- Led an 8-member team's post-conflict epidemiological baseline assessment for a 66,000-person catchment area — first comprehensive assessment of the site since the 2023–2025 conflict
- Designed and ran a 262-household survey (1,355 persons) using stratified random sampling; conducted 26 key informant interviews across 21 hospital staff
- Co-authored final report identifying critical gaps: workforce density at 24% of WHO minimum, 49.7% of OPD visits malaria-attributable, three disease outbreaks

TECHNICAL SKILLS

- **Data Science and Programming:** Python (pandas, NumPy, scikit-learn, SciPy, statsmodels, matplotlib, seaborn, plotly, GeoPandas), SQL, SPSS, Excel, Power BI, Jupyter, VS Code
- **Software Development:** HTML/CSS/PHP (web), Google Apps Script, Git and GitHub

LEADERSHIP AND SOFT SKILLS

- **Executive Leadership & Operations:** Crisis Leadership & Operations, Delegation & Succession Planning, Systems Design & Process Optimization
- **People & Resource Management:** Cross-Functional Team Building, Recruitment, Financial Stewardship under Scarcity
- **Communication & Engagement:** Stakeholder Management, Written and Verbal Communication
- **Adaptability & Mindset:** Resilience, Comfort with Ambiguity, Growth Mindset, Adaptability

CERTIFICATIONS

- IBM Data Science Professional Certificate · Coursera · February 2024 (10-course specialization)
- DeepLearning.AI Mathematics for Machine Learning and Data Science · Coursera · June 2025 (Linear Algebra, Calculus, Probability and Statistics)
- Google Project Management Professional Certificate · Coursera · August 2024
- IBM Data Science Fundamentals with Python and SQL · Coursera · October 2024
- MITx Introduction to Computer Science and Programming Using Python · edX · April 2025
- MITx Introduction to Computational Thinking and Data Science · edX · May 2025

- Certified Facilitator · International Federation of Medical Students Associations · May 2022 · 43 sessions, 100+ hours
- Pan African Youth Leadership Program · University of Texas at Austin · July 2017

LANGUAGES

Arabic — Native

English — Fluent (C1) · 600+ hours intensive instruction through US Embassy Access Program (2015–2017) · Professional use since 2015